

NIIS HACKATHON 2026 — OFFICIAL PROBLEM BRIEF

Track: AI & Machine Learning

Challenge ID: PS-09

Title: Explainable AI-Based Prediction & Decision Support System

1. Executive Summary & Problem Overview

Machine learning models are increasingly deployed to forecast high-stakes real-world outcomes—such as student academic failure risks, industrial machinery breakdowns, supply chain demand surges, fraudulent financial transactions, and hospital or campus resource allocation. However, the vast majority of high-accuracy models (such as deep neural networks and ensemble gradient-boosted trees) function as opaque "black boxes."

When an algorithmic model predicts that an industrial turbine will fail in 48 hours, or that a student is at high risk of dropping out, domain operators cannot afford to act blindly on a raw probability score. Without clear, auditable, and interpretable explanations detailing *why* a prediction was made, stakeholders face trust deficits, regulatory compliance bottlenecks, ethical risks, and costly false interventions.

The objective of **PS-09 (Explainable AI-Based Prediction & Decision Support System)** is to design and develop an end-to-end predictive machine learning solution for a high-impact operational domain. The platform must deliver accurate forecasts alongside transparent, human-understandable explanations—using local feature attributions, counterfactual "what-if" simulations, global feature importance charts, and natural-language executive action recommendations.

2. Core Operational Focus Areas (*Choose One Primary Domain*)

Participating teams must select and demonstrate their XAI system on a practical predictive problem, such as:

1. **Academic Performance & At-Risk Dropout Prediction:** Forecasting student course failure/dropout probability with actionable pedagogical remediation.
2. **Predictive Maintenance & Equipment Failure:** Forecasting industrial motor/sensor breakdown timelines with root-cause component attribution.
3. **Enterprise Demand & Supply Forecasting:** Predicting regional product/utility demand spikes to optimize warehouse inventories.
4. **Financial Fraud & Anomaly Detection:** Classifying suspicious transactions with explainable rule-and-feature collision evidence.

5. **Hospital / Campus Resource Utilization:** Forecasting bed occupancy, cafeteria meal demands, or emergency ward patient intake surges.

3. Core Personas & Access Roles

1. **Domain Decision-Maker (e.g., Academic Dean, Plant Manager, Risk Officer):** Reviews macro prediction dashboards, evaluates global risk drivers, and approves AI-recommended intervention policies.
2. **Operational Field Agent (e.g., Faculty Mentor, Maintenance Engineer, Fraud Analyst):** Inspects individual instance predictions, examines local feature attribution waterfalls, and reviews specific corrective steps.
3. **Data Science & ML Auditor:** Evaluates model performance metrics, audits algorithmic bias/fairness across subgroups, and verifies explainability consistency.
4. **End-User / Subject (e.g., Student, Customer, Worker):** Accesses a simplified, transparent explanation of factors influencing their score and self-improvement recommendations.

4. Key Functional Modules & Technical Deliverables

Module 1: Predictive Machine Learning Pipeline

- **Data Ingestion & Feature Engineering:** Robust preprocessing pipeline handling structured tabular, time-series, or event-log datasets with automated missing-value imputation and feature scaling.
- **Supervised Predictive Modeling:** Optimized baseline and advanced ML models (e.g., LightGBM, XGBoost, CatBoost, Random Forest, or Multilayer Perceptrons) delivering robust classification or regression accuracy.
- **Continuous Calibration & Metric Evaluation:** Transparent validation tracking precision, recall, ROC-AUC, F1-score, and root-mean-square error (RMSE) with k-fold cross-validation.

Module 2: Explainable AI (XAI) Attribution Engine

- **Local Feature Importance (SHAP / LIME):** Instance-level explanations showing exactly how specific variables pushed the prediction higher or lower (e.g., SHAP force plots, waterfall breakdowns).
- **Global Model Explainability:** Aggregate feature importance rankings revealing top systemic drivers across the entire dataset (e.g., SHAP summary beeswarm plots, partial dependence plots).
- **Feature Interaction Detection:** Identification of synergistic dependencies between multiple input variables (e.g., *"Low attendance (<65%) combined with mid-term score drop (<40%) increases failure risk by 3.8x"*).

Module 3: Counterfactual "What-If" Simulation Engine

- **Interactive Scenario Playground:** Enables operators to manually tweak feature inputs on an active prediction to view live output changes in real time.
- **Prescriptive Counterfactual Generation:** Algorithmic calculation of the minimal realistic feature adjustments required to reverse an adverse outcome (e.g., *"If Student improves lecture attendance from 62% to 75% and submits 2 pending assignments, Failure Risk drops from 84% to 28%"*).
- **Actionability Constraint Guardrails:** Ensures simulated recommendations only modify actionable attributes (preventing nonsensical tweaks to immutable traits like age, gender, or historical dates).

Module 4: Decision Support Console & Natural Language Reporting

- **Executive Summary Generator:** Converts statistical SHAP attributions into clear, professional, plain-language text summaries powered by LLM reasoning (Gemini API).
- **Automated Action Plan Builder:** Formulates prioritized operational checklists customized to the prediction severity level.
- **Exportable Audit Dossiers:** One-click generation of comprehensive, auditable PDF reports containing prediction confidence, data lineage, risk flags, and visual XAI charts.

5. Recommended Technology Stack

Layer, Suggested Technologies

Frontend Web / Console, React.js (Vite) / Next.js / Tailwind CSS / Recharts / Plotly.js / Lucide Icons

Backend API, Python (FastAPI / Flask) with RESTful JSON endpoints

Machine Learning & Core, "Python (Scikit-learn, XGBoost, LightGBM, CatBoost, Pandas, NumPy)"

Explainability Libraries, "SHAP (SHapley Additive exPlanations), LIME, InterpretML, DiCE (Diverse Counterfactual Explanations)"

Generative XAI Summaries, Google Gemini API (1.5 Flash/Pro) for plain-language reasoning briefs

Database & Cache, PostgreSQL / Supabase / SQLite + Redis for caching computed SHAP values

6. Evaluation & Judging Criteria (100 Points Total)

1. **Predictive Performance & ML Rigor (25 Points):** Robust model selection, proper validation splits, avoidance of data leakage, and high predictive accuracy metrics.
2. **Explainability Depth & Transparency (25 Points):** Meaningful, mathematically grounded local and global feature attribution (SHAP/LIME) rather than arbitrary rule-checking.
3. **Actionability & Counterfactual Utility (20 Points):** Real-world feasibility of the "what-if" simulator and the practical quality of prescriptive remediation steps.

4. **Dashboard UX & Decision Support Integration (15 Points):** Intuitive visualization of complex XAI plots, responsive design, and accessible natural language reporting.
5. **Sprint Execution, Code Quality & Documentation (15 Points):** Well-structured GitHub repo, reproducible sample dataset, clean .env.example, and an interactive live web demo.

7. Submission Guidelines

1. **Code Repository:** Public/Private GitHub repository containing clean branch commits, setup instructions, sample training/test datasets, and .env.example.
2. **Live Deployment:** Hosted interactive web application (Vercel / Render / Hugging Face Spaces / AWS).
3. **Demo Video / Deck:** A crisp 3-minute video showing:
 4. Ingestion of a sample data record.
 5. The prediction outcome with certainty confidence.
 6. Interactive exploration of the SHAP waterfall feature breakdown.
 7. Performing a live counterfactual "what-if" tweak to reverse an adverse prediction.
 8. Generating the natural-language executive decision brief.